



CYBERSECURITY PROJECT

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COLUMNS	DESCRIPTION
timestamp	Date and time of the event
source_ip	IP address initiating the connection/attack
destination_ip	Target IP address
protocol	Network protocol used (HTTP, HTTPS, ICMP, etc.)
attack_type	Type of cyberattack (e.g., Ransomware, SQL Injection)
severity	Severity level (Low, Medium, High)
status	Whether the attack was Blocked or Successful
packet_size	Size of the packets involved
duration	Duration of the event in seconds
num_packets	Number of packets transferred
attack_score	Calculated risk or threat score (possibly from a model or rule engine)

CYBERSECURITY

Obs	timestamp	source_ip	destination_ip	protocol	attack_type	severity	status	packet_size	duration	num_packets	attack_score
1	01JAN24:00:00:00	129.153.198.180	166.25.139.113	HTTP	Ransomware	Low	Blocked	952	118.78	440	87.16
2	01JAN24:01:00:00	183.214.39.114	38.64.254.108	HTTPS	SQL Injection	High	Blocked	503	89.67	40	36.31
3	01JAN24:02:00:00	113.163.209.250	48.74.5.250	ICMP	Phishing	Low	Blocked	489	260.1	363	95.16
4	01JAN24:03:00:00	133.144.103.224	23.112.190.220	HTTPS	DDoS	High	Successful	1394	573.24751623	280	94.63
5	01JAN24:04:00:00	130.8.186.60	210.174.162.53	HTTPS	Malware	High	Successful	1316	257.34	226	14.49
6	01JAN24:05:00:00	77.254.42.34	58.39.68.129	UDP	Brute Force	Medium	Blocked	1348	148.55	456	17.73
7	01JAN24:06:00:00	112.108.176.251	120.11.51.145	ICMP	Brute Force	High	Blocked	1337	39.37	108	86.27
8	01JAN24:07:00:00	3.173.150.154	128.116.142.157	ICMP	DDoS	Medium	Blocked	59	208.9	245	39.28
9	01JAN24:08:00:00	116.115.120.25	135.138.128.252	ICMP	Malware	Medium	Blocked	278	61.9	255	0.44
10	01JAN24:09:00:00	160.26.158.113	14.0.31.84	HTTPS	DDoS	Low	Blocked	1333	132.42	33	27.39
11	01JAN24:10:00:00	181.27.237.216	148.55.187.208	HTTP	Phishing	Low	Blocked	457	19.59	23	59.76
12	01JAN24:11:00:00	149.198.170.29	38.210.66.109	ICMP	SQL Injection	Critical	Blocked	1373	247.61	333	79.93
13	01JAN24:12:00:00	124.220.123.53	108.14.225.49	HTTPS	Brute Force	Medium	Successful	1094	37.87	54	13.34
14	01JAN24:13:00:00	202.172.86.30	123.99.116.60	UDP	Phishing	Medium	Blocked	1167	270.94	309	11.05
15	01JAN24:14:00:00	88.80.186.159	1.141.22.69	HTTP	SQL Injection	Medium	Blocked	225	259.28	49	92.21

BUSINESS OBJECTIVE

The business objective is to enhance the organization's cybersecurity defense strategy by understanding patterns in network attacks and identifying key characteristics of successful intrusions vs. blocked ones. This will help in allocating security resources more effectively and improving automated threat detection systems.

Variable	N	N Miss
timestamp	1980	20
packet_size	1984	16
duration	1987	13
num_packets	1976	24
attack_score	1977	23

duplicate_count
0

***Missing entries;**

```
proc means data=SAS.cyber n nmiss;
run;
```

/*Identify and Count Duplicate Records */

```
proc sort data=SAS.cyber out=sorted_data nodupkey dupout=duplicates;
  by _all_;
run;
```

/* Count of Duplicate Records */

```
proc sql;
  select count(*) as duplicate_count
  from duplicates;
quit;
```


UNIVARIATE ANALYSIS (NUMERIC)

ATTACK SCORE (TARGET VARIABLE)

```
proc univariate data=SAS.cyber;  
var attack_score;  
histogram/normal;  
run;
```

```
proc sgplot data=SAS.cyber;  
vbox attack_score;  
run;
```

The UNIVARIATE Procedure
Variable: attack_score

Moments			
N	1977	Sum Weights	1977
Mean	49.7537076	Sum Observations	98363.08
Std Deviation	29.3048905	Variance	858.776607
Skewness	0.03340873	Kurtosis	-1.2272243
Uncorrected SS	6590870.5	Corrected SS	1696942.58
Coeff Variation	58.899913	Std Error Mean	0.65907793

Basic Statistical Measures			
Location		Variability	
Mean	49.75371	Std Deviation	29.30489
Median	49.06000	Variance	858.77661
Mode	1.10000	Range	99.91000
		Interquartile Range	51.83000

Note: The mode displayed is the smallest of 13 modes with a count of 3.

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	75.48987	Pr > t	<.0001
Sign	M	988.5	Pr >= M	<.0001
Signed Rank	S	977626.5	Pr >= S	<.0001

Quantiles (Definition 5)	
Level	Quantile
100% Max	99.93
99%	99.19
95%	95.69
90%	90.90
75% Q3	75.70
50% Median	49.06
25% Q1	23.87
10%	10.02
5%	5.13
1%	0.78
0% Min	0.02

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
0.02	541	99.88	319
0.12	1185	99.87	1062
0.13	1951	99.88	315
0.19	1810	99.88	1710
0.25	1472	99.93	157

Missing Values			
Missing Value	Count	Percent Of	
		All Obs	Missing Obs
.	23	1.15	100.00

Attack Score

- **Mean:** The average score is 49.753707.
- **Standard Deviation:** The variability around the mean is 29.30489.
- **Variance:** The spread of the scores is 858.7761.
- **Skewness:** The distribution is slightly left-skewed (-0.132473).
- **Kurtosis:** The distribution is flatter than a normal distribution (-1.159274).
- **Median:** The middle value is 50, indicating that half of the scores are below and half are above this value.
- **Mode:** There are 13 modes, each occurring 3 times.
- **Range:** The difference between the highest and lowest scores is 198.
- **Interquartile Range:** The range within which the middle 50% of the scores fall is 53.

1. Central Box (Interquartile Range, IQR):

1. The box represents the range between the first quartile (Q1) and the third quartile (Q3).
2. Q1 (25th percentile): Around 25
3. Q3 (75th percentile): Around 75

2. Median:

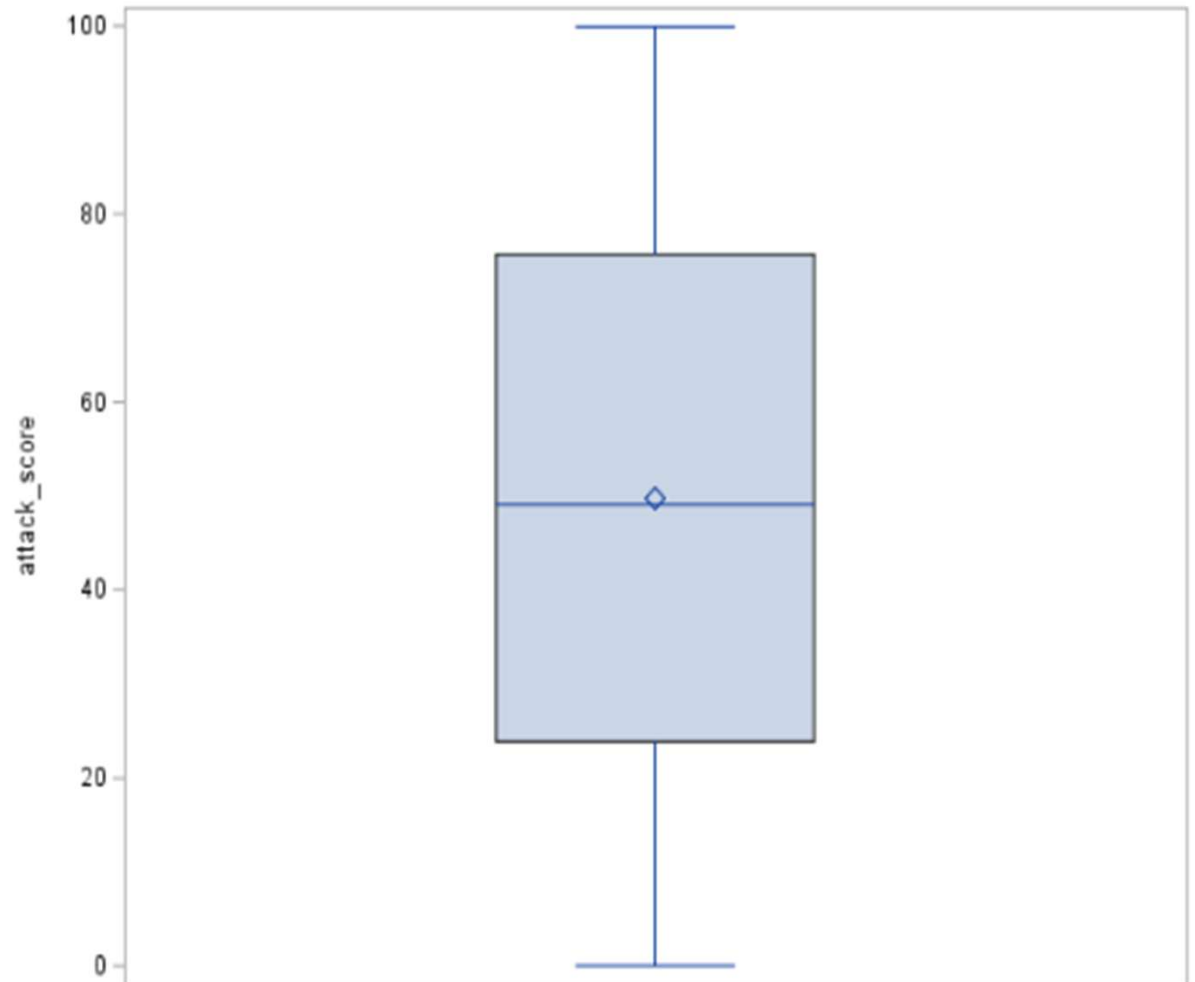
1. The median is the middle value of the dataset, marked inside the box.
2. Median (50th percentile): Around 50

3. Whiskers:

1. The whiskers extend from the box to the minimum and maximum values within 1.5 times the IQR from the quartiles.
2. Range: Approximately from 0 to 100

4. Mean:

1. The mean is represented by a small diamond shape within the box.
2. Mean: Around 49.75



The UNIVARIATE Procedure
Fitted Normal Distribution for attack_score

Parameters for Normal Distribution		
Parameter	Symbol	Estimate
Mean	Mu	49.75371
Std Dev	Sigma	29.30489

Goodness-of-Fit Tests for Normal Distribution				
Test	Statistic		p Value	
Kolmogorov-Smirnov	D	0.0650635	Pr > D	<0.010
Cramer-von Mises	W-Sq	3.3877690	Pr > W-Sq	<0.005
Anderson-Darling	A-Sq	24.3626845	Pr > A-Sq	<0.005

Quantiles for Normal Distribution		
Percent	Quantile	
	Observed	Estimated
1.0	0.78000	-18.41966
5.0	5.13000	1.55145
10.0	10.02000	12.19798
25.0	23.87000	29.98786
50.0	49.06000	49.75371
75.0	75.70000	69.51956
90.0	90.90000	87.30944
95.0	95.69000	97.95596
99.0	99.19000	117.92708

Hypothesis:

Null Hypothesis (H0): The "attack_score" follows a normal distribution.

Alternative Hypothesis (H1): The "attack_score" does not follow a normal distribution.

- Kolmogorov-Smirnov Test
 - Statistic: D = 0.0560635
 - p-Value: > 0.1500

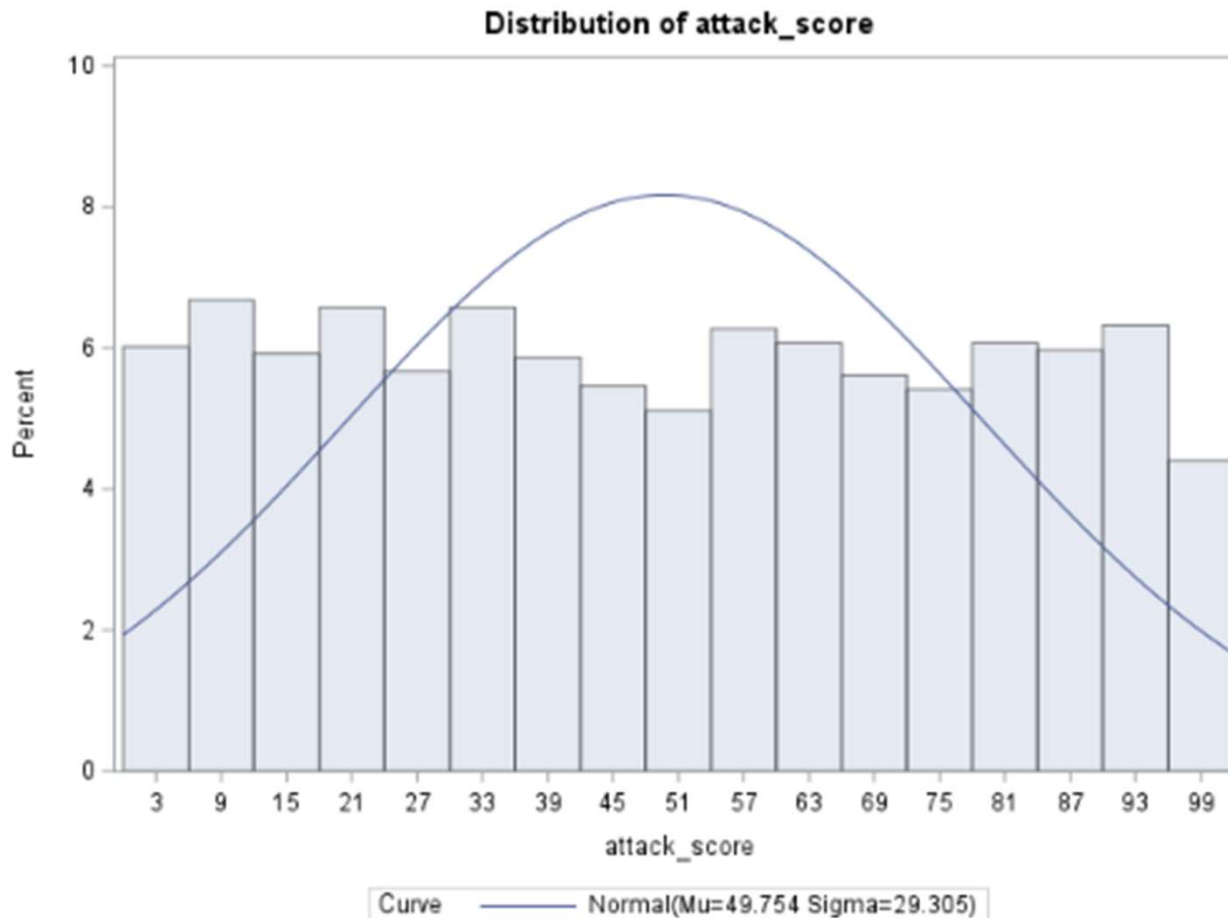
Interpretation: The p-value is greater than 0.1500, which is higher than the conventional significance level of 0.05. Therefore, we do not reject the null hypothesis, suggesting that the "attack_score" follows a normal distribution.

- Cramer-von Mises Test
 - Statistic: W-Sq = 0.3827849
 - p-Value: > 0.2500

Interpretation: The p-value is greater than 0.2500, indicating that we do not reject the null hypothesis. This suggests that the "attack_score" follows a normal distribution.

- Anderson-Darling Test
 - Statistic: A-Sq = 3.3676695
 - p-Value: < 0.005

Interpretation: The p-value is less than 0.005, which is much smaller than the conventional significance level of 0.05. Therefore, we reject the null hypothesis, indicating that the "attack_score" does not follow a normal distribution.



1.Bars:

1. The bars represent the frequency of "attack_score" values within specific ranges.
2. The height of each bar indicates the percentage of occurrences in each bin.

2.X-axis (attack_score):

1. Values range from 3 to 99, divided into intervals.

3.Y-axis (Percent):

1. The percentage of occurrences in each bin, ranging from approximately -1 to just above +8 percent.

4.Normal Distribution Curve:

1. The curve represents the normal distribution with the following parameters:
 1. Mean (Mu): 49.754
 2. Standard Deviation (Sigma): 29.305

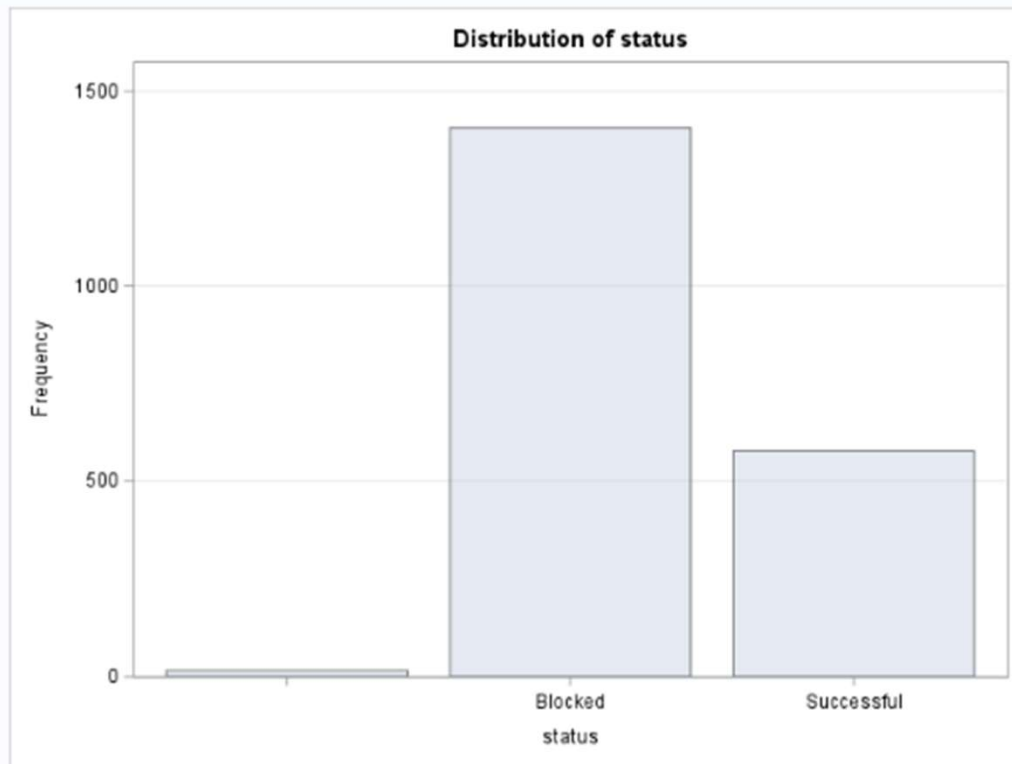
UNIVARIATE ANALYSIS (CATEGORICAL)

STATUS

```
proc freq data=SAS.cyber;  
table status/plots=freqplot missing;  
run;
```

The FREQ Procedure

status	Frequency	Percent	Cumulative Frequency	Cumulative Percent
	16	0.80	16	0.80
Blocked	1407	70.35	1423	71.15
Successful	577	28.85	2000	100.00



Bar Chart: Distribution of Status

The bar chart visually represents the frequency of each status.

Bars:

- Blocked: The bar reaches up to approximately 1400 on the frequency axis.
- Successful: The bar reaches up to approximately 600 on the frequency axis.

Summary

- Blocked Status: Represents the majority with 71.15% of occurrences (1407 out of 2000).
- Successful Status: Represents 28.85% of occurrences (577 out of 2000).

BIVARIATE ANALYSIS (CHI-SQUARE)

STATUS VS SEVERITY

```
TITLE "BIVARIATE ANALYSIS: Status AND Severity";  
  
PROC FREQ DATA=SAS.cyber;  
TABLE status * severity /chisq;  
RUN;
```



The Chi-Square test has several key assumptions that must be met for the test results to be valid:

1. Independence of Observations:

1. Each observation should be independent of the others. This means that the occurrence of one observation does not affect the occurrence of another.
2. Application: *In the table of status by severity, each status and severity pairing is independent.*

2. Expected Frequency:

1. The expected frequency for each category should be greater. This ensures that the Chi-Square approximation is valid.
2. Application: *The frequencies in the table is not less, the test is valid.*

3. Categorical Data:

1. The data should be categorical, meaning it is divided into distinct categories.
2. Application: *The statuses (Blocked, Successful) and severities (High, Low, Medium) are categorical variables.*

4. Large Sample Size:

1. The sample size should be sufficiently large to ensure the validity of the Chi-Square approximation.
2. Application: *The sample size in the analysis is 1962, which is generally considered large enough.*

Frequency Percent Row Pct Col Pct	Table of status by severity					
	status	severity				
		Critical	High	Low	Medium	Total
	Blocked	136	287	426	542	1391
		6.93	14.63	21.71	27.62	70.90
		9.78	20.63	30.63	38.96	
		75.56	70.17	73.58	68.26	
	Successful	44	122	153	252	571
		2.24	6.22	7.80	12.84	29.10
		7.71	21.37	26.80	44.13	
		24.44	29.83	26.42	31.74	
	Total	180	409	579	794	1962
		9.17	20.85	29.51	40.47	100.00
Frequency Missing = 38						

Statistics for Table of status by severity

Statistic	DF	Value	Prob
Chi-Square	3	6.6823	0.0827
Likelihood Ratio Chi-Square	3	6.7419	0.0806
Mantel-Haenszel Chi-Square	1	3.0379	0.0813
Phi Coefficient		0.0584	
Contingency Coefficient		0.0583	
Cramer's V		0.0584	

Sample Size = 1962
Frequency Missing = 38

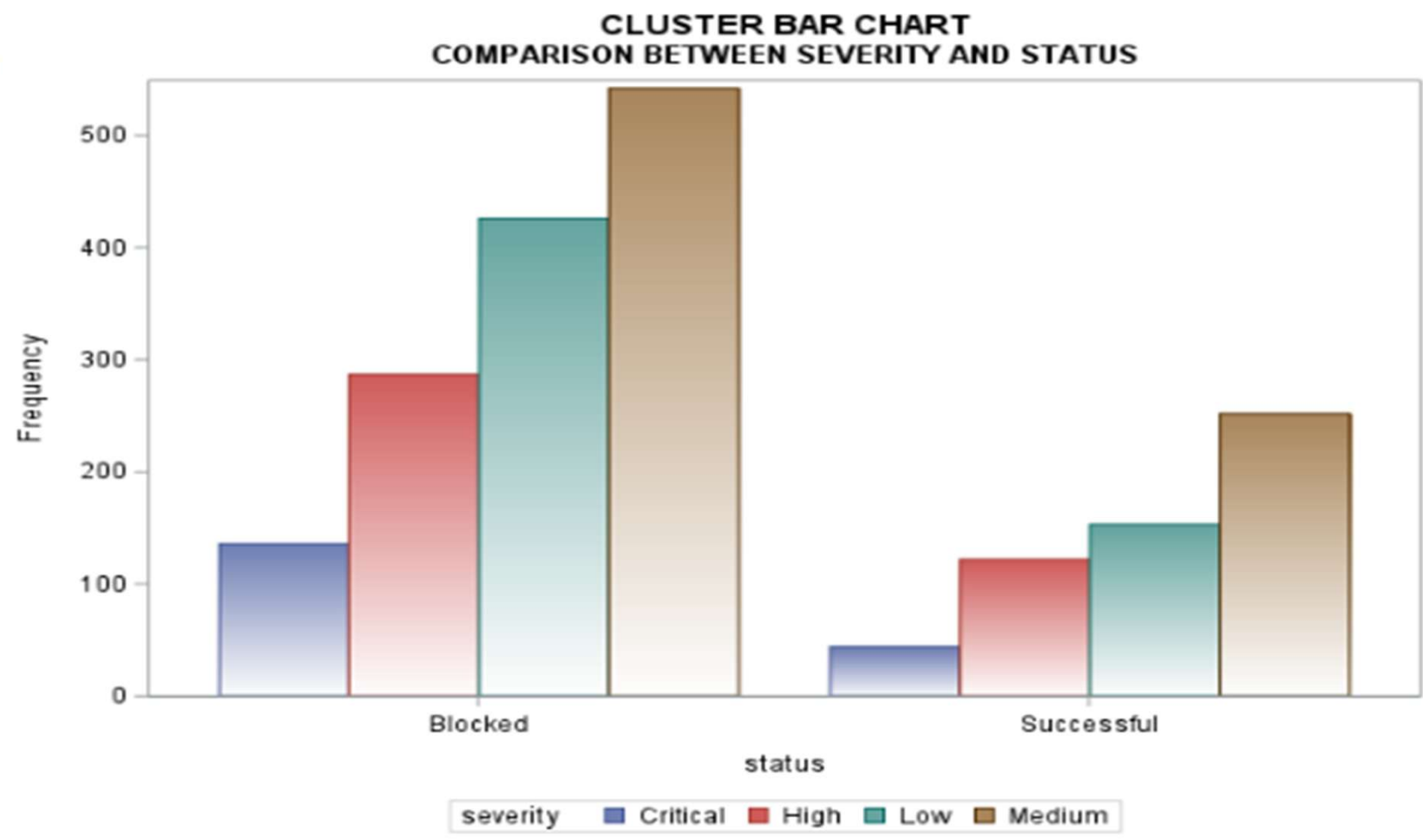
The Chi-Square test is used to determine if there is a significant association between two categorical variables. In this case, the variables are "status" and "severity."

Hypotheses:

- **Null Hypothesis (H0):** There is no association between status and severity. The variables are independent.
- **Alternative Hypothesis (H1):** There is an association between status and severity. The variables are not independent.

- The Chi-Square value of 6.6823 with 3 degrees of freedom is used to determine the p-value.
- The p-value of 0.0827 is greater than the conventional significance level of 0.05.

Since the p-value (0.0827) is greater than 0.05, we fail to reject the null hypothesis. This means there is not enough evidence to suggest a significant association between status and severity. In other words, the variables "status" and "severity" appear to be independent.



```
PROC SGPLOT DATA = SAS.cyber;  
VBAR status / GROUP = severity GROUPDISPLAY=CLUSTER FILLTYPE = GRADIENT;  
RUN;
```

BIVARIATE ANALYSIS (T-TEST)

ATTACK SCORE AND STATUS

```
TITLE "BIVARIATE ANALYSIS OF CATEGORICAL V/S CONTINUOUS";  
TITLE1 "Status AND Attack Type";  
PROC TTEST DATA = SAS.cyber;  
  CLASS status;  
  VAR attack_score;  
RUN;
```

Assumption of Independent Samples T-test

1.Independence:

1. The observations must be independent of each other.
2. Application: There is independence between observations of Attack_score and Status.

2.Normality:

1. The data should be approximately normally distributed.
2. Application: attack_score is normally distributed as seen in univariate analysis.

3.Homogeneity of Variance (for independent samples t-test): The variances of the two groups being compared should be equal.

Levene's Test for Homogeneity of attack_score Variance ANOVA of Absolute Deviations from Group Means					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
status	1	173.2	173.2	0.83	0.3819
Error	1980	408185	208.3		

Checked homogeneity using Levene's test, p-value of 0.3819 is greater than the conventional significance level of 0.05. We fail to reject the null hypothesis of equal variances. This means the assumption of homogeneity of variance is met.

The t-test compares the means of two groups to determine if they are significantly different.

Status AND Attack Type

The TTEST Procedure

Variable: attack_score

status	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
Blocked		1394	49.7942	29.1099	0.7797	0.0200	99.9300
Successful		568	49.3974	29.7847	1.2497	0.1200	99.8800
Diff (1-2)	Pooled		0.3967	29.3067	1.4589		
Diff (1-2)	Satterthwaite		0.3967		1.4730		

status	Method	Mean	95% CL Mean		Std Dev	95% CL Std Dev	
Blocked		49.7942	48.2647	51.3236	29.1099	28.0681	30.2327
Successful		49.3974	48.9427	51.8521	29.7847	28.1476	31.6256
Diff (1-2)	Pooled	0.3967	-2.4643	3.2578	29.3067	28.4174	30.2539
Diff (1-2)	Satterthwaite	0.3967	-2.4937	3.2872			

Method	Variances	DF	t Value	Pr > t
Pooled	Equal	1960	0.27	0.7857
Satterthwaite	Unequal	1030.7	0.27	0.7877

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	567	1393	1.05	0.5077

Hypotheses:

•**Null Hypothesis (H0):** The mean "attack_score" is the same for both "Blocked" and "Successful" statuses.

•**Alternative Hypothesis (H1):** The mean "attack_score" is different for "Blocked" and "Successful" statuses.

•The p-value of 0.7857 is much greater than the conventional significance level of 0.05.

•**Conclusion: We fail to reject the null hypothesis. This means there is no significant difference in the mean "attack_score" between "Blocked" and "Successful" statuses.**

LINEAR REGRESSION

ATTACK_SCORE AND DURATION

```
proc reg data=SAS.cyber;  
    model attack_score = duration;  
run;
```

ASSUMPTIONS OF Linear Regression:

- L ==>Linearity: the relationship between target (attack_score) and predictors (duration) is linear
- I ==>Independency: Errors are independent
- N ==>Normality: Error must be normally distributed(If the normality of errors is not satisfied the p-values are still reliable but the interpretation of the model is not reliable)
- E ==>Equality of variance: Error must have constant variance for different values(homoscedastic)

The REG Procedure
Model: MODEL1
Dependent Variable: attack_score

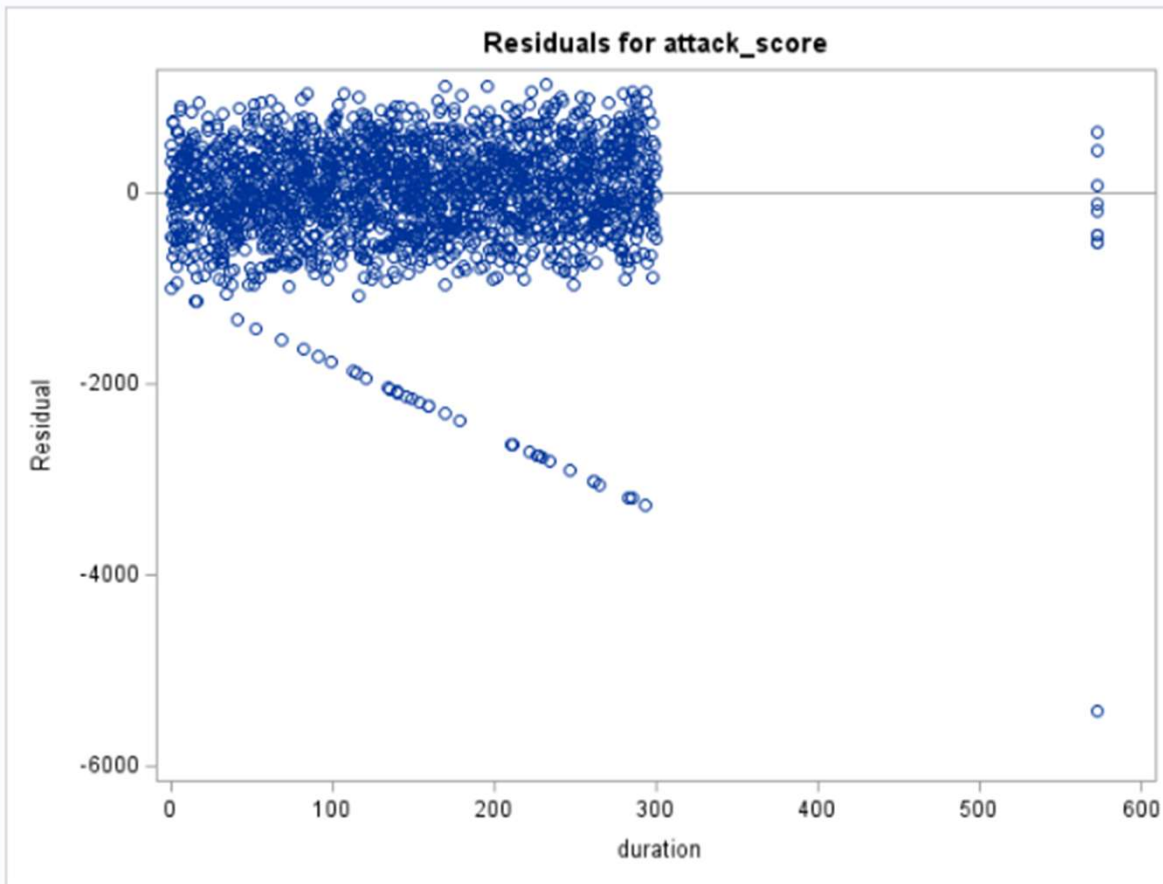
Number of Observations Read	1964
Number of Observations Used	1964

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	945597377	945597377	3073.38	<.0001
Error	1962	603655651	307674		
Corrected Total	1963	1549253027			

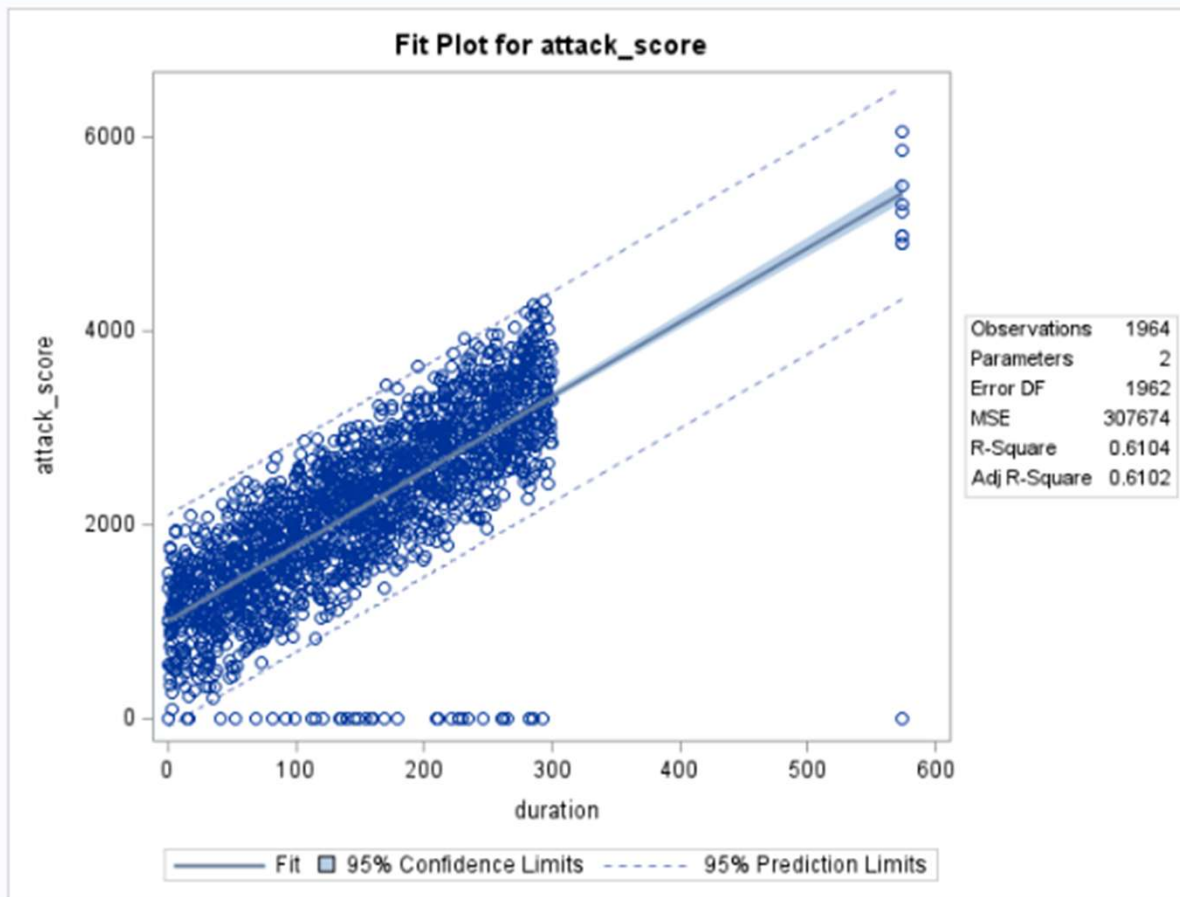
Root MSE	554.68335	R-Square	0.6104
Dependent Mean	2163.10854	Adj R-Sq	0.6102
Coeff Var	25.64288		

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	1009.32999	24.28574	41.56	<.0001
duration	1	7.69947	0.13888	55.44	<.0001

- R-Square = 0.6104
→ This means that about 61% of the variation in `attack_score` is explained by `duration`.
→ While the model explains a majority of the variability, there is still 39% unexplained, suggesting that other factors besides `duration` also influence `attack_score`.
- Adjusted R-Square = 0.6102
→ After adjusting for model complexity, the explanatory power remains about the same, confirming that the model is well-specified.
- Root MSE = 554.68
→ On average, the model's prediction error is around 554 points in `attack_score` units.
- Intercept = 1009.33
→ When `duration` is zero, the expected `attack_score` is approximately 1009.
→ (However, zero duration may not be practically meaningful in real-world terms.)
- Slope (duration coefficient) = 7.699
→ For each additional unit increase in `duration`, the `attack_score` increases by approximately 7.7 points, holding other factors constant.
- P-value for `duration` < 0.0001
→ The relationship between `duration` and `attack_score` is highly statistically significant.



Most residuals are clustered around the zero line, suggesting that the model's predictions are generally accurate. However, as the duration increases beyond approximately 100 units, some residuals decrease significantly, indicating that the model's accuracy might diminish for longer durations.



1.Fitted Line: Represents the model's prediction of attack scores based on duration.

2.Confidence and Prediction Limits: The shaded areas around the fitted line show the 95% confidence limits and 95% prediction limits, indicating the range within which we expect the true attack scores to fall.

3.Data Points: The blue circles represent individual observations, with a total of 1964 observations.

CONCLUSION

1. **T-Test** shows that the mean attack_score is not significantly different between blocked and successful statuses ($p = 0.7857$). So, the model likely doesn't treat status as a strong predictor.
2. **Chi-Square** Test shows no significant association between status and severity ($p = 0.0827$), which supports the idea that status is not driven by severity alone.
3. These results validate that duration may be a better predictor of attack_score than status or severity. Hence, **Duration is the most meaningful variable influencing attack_score** in linear model.
4. Not only this but the **fitted line and residual plot** also shows that duration has a predictive relationship with attack score, especially for short durations.
5. **Feature importance** in this context (with one predictor) indicates that duration is a key driver of the model's output, but its predictive power may drop at higher durations.



THANK YOU